



Bhargavi Poyekar

in /bhargavi-poyekar ✉ poyekarbhargavi@gmail.com 📞 9930945072 📍 Mumbai, MH

CORE COMPETENCIES

- Competent in coding, testing and debugging. Noteworthy abilities include:
 - Ansible and Terraform** for automation of infrastructure deployment and management. 🔗
 - Python** Distributed System for Car booking. 🔗
 - Flask** API for calculating points from a receipt using **Docker** environment and **Github Actions**. 🔗
- Certified as **AWS DVA-C02**, **AWS CLF-C02**, and **Terraform Associate (003)**.
- Proficiency in **Javascript**, **Ajax**, **MySQL**, **Linux**, and **Git** for full stack development and version control.
- Strong understanding of software testing methodologies, including **manual**, **unit**, and **integration testing**.
- Exceptional communication, interpersonal & analytical problem solving skills.

PROFESSIONAL EXPERIENCE

Software Developer, University of Maryland, Baltimore County Aug 2024 – June 2025

- Built scalable backend infrastructure to provision virtual lab environments for 50+ students using **Terraform**, **Ansible**, and **vCenter APIs**, enabling secure multi-VM setups with internal, external, and DMZ network segmentation.
- Programmed modular **Infrastructure-as-Code** components and automated deployment scripts, accelerating environment setup by **90%** and improving cross-course reusability.
- Streamlined VM provisioning via backend logic integrated with web-based dashboards, enhancing instructor control and reducing manual intervention.

Software Engineer (Cloud) Intern, University of Maryland, Baltimore County Jun 2023 – May 2024

- Engineered a secure, full-stack **Django web application** for managing 50+ Linux VMs, integrating **role-based access control (RBAC)**, **SSL/TLS**, and secure session handling to support 100+ users.
- Automated VM lifecycle operations using **Python** and **Ansible**, embedding **SSH hardening**, jump host routing, and key-based authentication to enhance system security and reduce manual overhead.
- Utilized **Ansible Vault** for secure credential storage, encrypting sensitive data and ensuring controlled access for over 100 students and faculty using the VM infrastructure.

Software Developer Intern, Origin Cloud Tech Jun 2020 - Dec 2020

- Developed a **scalable and modular** LMS platform using **Laravel MVC**, improving **user engagement by 2%** for 10K+ monthly users. Strengthened system security with **OAuth2** and **multi-factor authentication**, and integrated **REST APIs** for payments (PayPal, Razorpay), geolocation, and Zoom, boosting website efficiency by **40%**.
- Crafted a **responsive and accessible** UI with **HTML5**, **CSS3**, **Bootstrap**, and **JavaScript**, reducing bounce rates by **15%** via loading feedback (skeleton screens, spinners). Managed deployments via CI/CD pipelines (GitHub Actions).
- Designed a database architecture with **normalized** tables, **multi-table joins**, and optimized **SQL CRUD** queries, reducing query execution time by **10%**. Wrote automated tests using **PHPUnit** to ensure data integrity.
- Collaborated in **pair programming** sessions with senior developers, reinforcing **OOP** and **SOLID** principles.

PROJECTS

HDFS to S3 Secure Data Transfer Application (Python, Hadoop, AWS):

- Developed a **Python** application to automate file transfers from **Hadoop HDFS** to **AWS S3**, reducing manual data handling time by **50%**. Implemented **IAM Role policies**, **STS AssumeRole** for temporary security credentials, and **client-side encryption**. Enhanced security by using **AWS SSM Parameter Store** to store sensitive credentials, replacing environment variables.
- Integrated **Amazon CloudWatch** to monitor file transfers, track errors, and log performance metrics. Configured **Amazon SNS** to send email notifications for transfer status updates (success/failure).
- Tech:** HDFS, S3 Bucket, IAM Roles, Python, boto3, GIT, Linux, CloudWatch, SNS, STS, SSM.

Flask API for Receipt Processing (Containerized, Scalable Deployment, SOLID Principles):

- Delivered a **Flask-based API** for calculating reward points, using **Docker** to ensure consistency across environments.
- Implemented **API testing** using **Postman**, automating tests for different API endpoints. Integrated continuous testing into **GitHub Actions** for automated deployment and validation.
- Tech:** Python, Flask, JSON, Docker, Docker Compose, Postman, GitHub Actions, Pytest.

Task Management API (JWT Authentication, RESTful Design, Rate Limiting):

- Established a secure **REST API** architecture for task management with **JWT token enforcement**.
- Enforced **API best practices** including versioning, consistent JSON responses, error handling, and rate limiting while ensuring **100%** test coverage through unit and integration tests.
- Tech:** Django REST Framework, JWT, SQLite, pytest, Postman, GitHub Actions, TLS, HTTPS.

TECHNICAL SKILLS

Languages and Databases: Python, Java, C/C++, PHP, Go, JavaScript, SQL, HTML, CSS, MySQL, SQLite, MongoDB.
Frameworks: Django, Django-Rest, Laravel, Selenium, Node.js, React, Flutter, Android Studio, Scikit, TensorFlow.
DevOps and Cloud: Ansible, Terraform, AWS, Docker, Kubernetes, EC2, S3, Lambda, IAM, VPC, Route53, Vault.
Others: GIT, Linux, VSCode, Pytest, Matplotlib, Tableau, jQuery, Pandas, Numpy, Rest API, Ajax, Postman.

EDUCATION

University of Maryland, Baltimore County - Baltimore, MD | MS in Computer Science | GPA: 3.85/4.0
Sardar Patel Institute of Technology, Mumbai, India | B.Tech in Computer Engineering | GPA: 9.4/10

Aug 2022 - May 2024
Aug 2018 - May 2022